Technology Stack

CrimeVision AI — Intelligent Multimodal Forensic Analysis Platform

Date	29 October 2026	**Team ID**	Not applicable — solo project
Project Name	CrimeVision AI	**Maximum Marks**	2 Marks (template criterion)

Define Your Technology Stack

The proposed stack combines a Python-based backend with computer-vision, audio-transcription, document-extraction, and language-model components. The dashboard presents source-linked candidate findings for investigator review. Technologies marked **To be decided** are intentionally left open for confirmation during implementation.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	**HTML5, CSS3, JavaScript** Three.js for optional 3D visualization threejs.org	Build a straightforward investigator dashboard for evidence upload, case navigation, analysis status, timelines, findings, and report viewing. Three.js is reserved for 3D views if they add value.
2	Backend / Server-Side	**Python 3.9+** Flask (Backend API) python.org · Flask documentation	Python supports the AI/ML ecosystem and ties the analysis components together. Flask provides HTTP endpoints for uploads, processing requests, case data, and results.
3	Database / Data Storage	**To be decided** Prototype file storage + structured metadata	Keep original evidence separate from derived outputs. Store case metadata, evidence references, processing status, timestamps, and model outputs. Select a persistent database after the data model and deployment needs are confirmed.
4	Cloud / Hosting / Deployment	**Local development initially** Production host: To be decided	Local execution supports iterative development and model testing. Hosting should be selected after estimating model memory, GPU needs, evidence sensitivity, API costs, and access-control requirements.
5	Version Control & CI/CD	**Git** Remote repository / CI service: To be decided	Git tracks source changes and supports safe experimentation. A remote repository and automated testing/deployment workflow can be selected later; secrets, API keys, evidence files, and private case data must not be committed.
6	Third-Party APIs / Other Tools	**Groq API** Visual Studio Code code.visualstudio.com	The Groq-hosted LLM is intended for evidence-grounded classification, reasoning, summaries, and investigator chat. VS Code is the primary development environment. External API use must be documented because selected text or metadata may leave the local machine.

AI, Media & Evidence Processing Components

Component	Technology	Purpose
Image / video processing	**OpenCV** · opencv.org	Decode and process supported images/video, extract frames, and prepare inputs for detection or captioning.
Model inference	**PyTorch** · pytorch.org	Run compatible model inference; exact model versions and hardware acceleration will be confirmed during setup.
Object detection	**RT-DETR**	Detect candidate objects in supported visual evidence. Model source/version and license to be verified before inclusion.
Image captioning	**BLIP / GIT / Florence alternative**	Generate image descriptions as supplementary context; final model selection depends on quality, license, memory use, and local compatibility.
Audio transcription	**Whisper**	Convert supported audio to text and preserve timestamps where available.
Document extraction	**OCR / text extraction tool — To be decided**	Extract searchable text from supported documents or scanned pages. Select the library based on PDF/image needs and evaluation.

Items intentionally left open: database engine, hosting/deployment target, remote Git/CI provider, OCR library, exact RT-DETR and captioning model versions, and production authentication approach. These are not assumed to be finalized.

Security and forensic integrity: preserve original evidence unchanged; associate each derived result with its source file and available frame/page/timestamp; keep API keys in environment variables; restrict access to case material; and label AI-generated observations as candidate leads requiring human review. Do not use live law-enforcement or biometric databases in the prototype.